

What Happens Next? Closed- and Open-World Video Action Recognition on a 33-Class Something-Something V2 Subset

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Abstract

We present a two-track study of temporal action prediction on a 33-class subset of Something-Something V2 (SSv2) [7], “What Happens Next?”: only the first $\approx 40\%$ of each clip (four JPEG frames) is given, so the outcome is withheld and classification must reason about motion. In **Track A (Closed World)**, where pretraining is forbidden, two data-centric fixes dominate architecture: a 10-epoch regularisation warm-up (+36.3pp) and matching the model’s frame count to the true clip length (+4.4pp, beating a $3\times$ parameter scale-up). A log-softmax ensemble of Temporal-Shift-Module [13] and CNN-Transformer models with test-time augmentation reaches **44.03%** top-1 on the official validation set. In **Track B (Open World)**, we fine-tune V-JEPA 2 [2] and VideoMAE [18] backbones with progressive unfreezing, layer-wise learning-rate decay, window-capped source-frame re-extraction and pseudo-labelling, then build a 14-model uniform ensemble reaching **0.6586** Kaggle top-1 (16th in both tracks). We document a self-discovered data-leakage event (a discarded 0.81 submission) and show that ensemble diversity across architecture families beats single-family or learned-weight ensembling. Failure analysis attributes the residual errors to real-vs-pretended and direction-ambiguous classes that four frames under-determine.

1. Problem Description

Task. The challenge classifies four-frame JPEG clips into one of 33 motion-defined action classes (e.g. *Pouring something into something*, *Pulling something from left to right*). Crucially, each clip contains only the *anticipatory* portion of the source SSv2 video—the shipped `frame_003` sits at a median fraction of 0.40 of the source clip—so the action’s outcome is deliberately withheld and the task probes motion prediction, not final-state recognition [7].

Dataset. The split provides 44,993 training clips, 6,745 validation clips and 6,913 test clips over 33 classes (numeric prefixes 000–032; class 027 is absent from training). Classes are heavily imbalanced: the largest (009, *Moving something up*) has 3,170 examples while the smallest (026, *Spilling something next to something*) has only 162, a ratio of $19.6\times$. The KL divergence between training and validation class distributions is 0.088, confirming the splits are representative.

Evaluation and tracks. Submissions are scored by top-1 accuracy on a public+private Kaggle leaderboard over two stages (Stage 1, intermediate; Stage 2, final); top-5 is tracked internally. The provided baseline reaches ≈ 0.55 private and the second-place public reference is 0.74; our final submissions placed **16th in both tracks** (counting the multiple accounts fielded by some teams). Following the challenge, we report two regimes. **Track A (Closed World)** prohibits pretrained models and external data—all spatial features, temporal dynamics and class boundaries are learned from scratch on the four provided frames. **Track B (Open World)** permits pretrained backbones and external corpora but not future frames or test labels. The task is hard for three reasons: (i) labels are motion-defined, defeating appearance-only models [7]; (ii) four frames is an order of magnitude below the 8–32 standard in video recognition [3, 13]; (iii) the discriminative cue—the action’s outcome—is withheld by construction.

2. Methods

2.1. Track A — Closed World

The pipeline evolved through six iterations on an internal training-split, all trained from scratch. Our guiding hypothesis is that, without pretraining, *every parameter spent on temporal modelling is one not spent learning spatial features*, favouring parameter-free temporal mechanisms over heavy temporal heads.

2.1.1. Architecture family

A ResNet-18 [8] with temporal average-pooling (**CNNBaseline**) and a per-frame-feature **CNN-LSTM** serve as lower bounds; with four frames the LSTM gives only marginal gains because backpropagation through time on such short sequences is unstable. **VideoFormer-Lite (VFL)** passes per-frame ResNet-18 features and a learnable [CLS] token [6] through a 2-layer pre-norm Transformer encoder [21] (global temporal attention over the four tokens) and classifies the [CLS] representation, reaching 53.37% internal; it is a lightweight relative of TimeSformer [3] that keeps a CNN spatial encoder for sample-efficiency from scratch. **TSM-ResNet** [13] wraps every residual block of ResNet-18: $1/\text{fold_div}$ of channels are shifted one step backward and another $1/\text{fold_div}$ one step forward along the time axis, mixing temporal information at *zero parameter cost* by re-indexing tensor slices. TSM was designed for SSv2-style motion labels and is therefore a strong prior here (Fig. 1); the two architectures’ biases are deliberately orthogonal — local-shift mixing in TSM vs. global attention in VFL — which drives the ensemble gain (Sec. 3).

2.1.2. Critical frame-count fix

The single largest gain (+4.4pp) came from aligning the `num_frames` parameter with the true clip length. Clips contain exactly four real frames; `num_frames > 4` triggers `linspace` interpolation that duplicates frames and corrupts the temporal-shift signal TSM relies on. Setting `num_frames = 4` brought `tsm_ultra_v2` to 57.73% internal; scaling instead to ResNet-50 with $T=8$ regressed to 45.26% (Fig. 2, Fig. 8), confirming that capacity cannot compensate for corrupted temporal input — fix the data, then scale.

2.1.3. Regularisation recipe

All Track A models share a regularisation stack (Table 1) whose most critical element is a *warm-up delay* on MixUp [23] and CutMix [22]: both are disabled for the first 10 epochs so the backbone can form basic per-frame representations before label-mixing is applied — a single change worth +36.3pp on VFL ($9.45 \rightarrow 45.78\%$). The intuition: a randomly-initialised classification head cannot meaningfully mix labels whose images it has not yet learned to distinguish, so early MixUp injects pure noise into the loss and collapses representations. The stack also includes label smoothing [17], RandAugment [5], random erasing [24], stochastic depth [11], and AdamW [16] with cosine annealing [15]. Focal loss [14] ($\gamma=2$) concentrates gradient on confusable class pairs — specifically the real-vs-pretended sinks (Sec. 3.3). Augmentation is domain-aware: *horizontal flip is disabled* because it inverts SSv2 direction-encoded labels (e.g. *Pulling left $\rightarrow$ right*); CutMix probability is also kept low ($p=0.25$) because a cut patch spans all four frames

Technique	Setting	Source
Reg. warm-up	10 ep, no MixUp/CutMix	this work
MixUp / CutMix	$\alpha=0.2$ / $p=0.25$, after ep 10	[22, 23]
Focal loss	$\gamma=2$	[14]
Label smoothing	$\varepsilon=0.1$	[17]
RandAugment	2 ops, mag. 0.5	[5]
Random erasing	$p=0.25$	[24]
Horizontal flip	disabled	label semantics
Stochastic depth	DropPath 0.1	[11]
Optimiser	AdamW, lr 10^{-3} , cosine	[16]

Table 1. Track A training recipe (best single model: TSM-ResNet18, $T=4$, `fold_div=4`, 150 epochs, ~ 5.8 h on one RTX A4000).

and frequently occludes the hand/object motion that defines the label.

2.1.4. Rotating k -fold and ensemble

Beyond the standard train/val split, the strongest Track A checkpoints have a *rotating-fold* variant that is retrained on every fold of the train+val union, providing higher-bias snapshots that nonetheless contribute uncorrelated errors at inference (Fig. 7). The final submission fuses three checkpoints by log-softmax late fusion with 10-crop spatial TTA (5 crops $\times$ 2 flips): `tsm_ultra_v2` (57.73% internal), its rotating-fold variant (`tsm_ultra_v2_rot`, trained on all folds), and `video_former_lite_ultra_rot`. Averaging in log-probability space down-weights over-confident errors and rewards agreement, suiting members of different calibration. The ensemble reaches **44.03% on the official val_dir** (Fig. 3); 10-crop TTA alone contributes ~ 1 pp on the strongest single model (Fig. 5), with a visible trade-off on direction-encoded classes (Fig. 6) where the 5 hflip views push some confidences toward the mirror class even after label-aware logit remapping.

2.2. Track B — Open World

Top SSv2 teams use video transformers pretrained on large unlabelled corpora. We selected two self-supervised families: **VideoMAE** [18] (masked-autoencoder pre-training, building on image MAE [9]/BEiT [1]; HuggingFace Kinetics-400 checkpoints) and **V-JEPA 2** [2], a joint-embedding predictive architecture that reconstructs masked regions in *feature* space rather than pixel space, focusing capacity on semantically relevant motion. We use the self-supervised `vjepa2-vitl-fpc64-256` checkpoint, pretrained on unlabelled video without any SSv2 labels—a requirement for integrity (Sec. 3.4).

Progressive unfreezing with LLRD. We fine-tune V-JEPA in three phases with layer-wise learning-rate decay (LLRD) [1, 4, 10], where lower encoder block ℓ receives

$\eta \cdot \lambda^\ell$ with $\lambda=0.75$. *Phase 0*: a frozen attentive probe (head only) reaches 0.672 EMA internal val. *Phase 1*: unfreezing the top-6 blocks with LLRD reaches 0.705. *Phase 2*: full-encoder fine-tuning reaches 0.728 internal and 0.6411 val_dir. Freezing first stabilises the head and preserves the pretrained feature geometry before large gradients reach lower layers; the characteristic staircase — each phase warm-starting above the previous plateau — is visible in Fig. 10. (These training phases are distinct from the challenge’s Stage 1/Stage 2 leaderboard milestones.)

Window-capped source-frame re-extraction. V-JEPA benefits from 16 input frames, but only four are shipped. Because folder names encode SSv2 video IDs, the full source WebM files are accessible. We re-extract 16 frames per clip in a *window-capped* manner: the shipped frame_003 is matched against source frames by pixel-level mean absolute error to locate the end of the provided window, 16 frames are sampled uniformly within $[0, \text{window_end}]$, and the search is hard-bounded to the first 50% of the source clip so the withheld outcome ($> 60\%$) is unreachable by construction (median window fraction 0.39, max 0.50; Fig. 11).

Pseudo-labelling. Following noisy-student self-training [12, 20], the Phase-2 snapshot ensemble generated soft labels for the test clips; predictions with max probability ≥ 0.85 were added as pseudo-labels (the model’s over-confident calibration meant that this threshold retained all 6,913 test clips, so pseudo-labelling effectively augments the train set with the entire test distribution under V-JEPA-ft’s own predictions). The model is then continue-finetuned on $\text{train} \cup \text{pseudo-test}$. The resulting V-JEPA-pseudo reaches 0.6519 val_dir / 0.6410 Kaggle, but the widened val/Kaggle gap (Sec. 3) signalled overfitting to the validation distribution, so pseudo-labelling was not iterated.

Honest cross-preprocessing ensemble. The final Track B system aggregates 14 checkpoints from three families, each fine-tuned with its own preprocessing: V-JEPA (6 snapshots, 16-frame 256px window-capped), VideoMAE-Base K400 (3, 4-frame 224px), VideoMAE-Large K400 (3, 4-frame 224px) and TSM-ResNet50 (2, 4-frame). Inference runs each model on its own preprocessing chain; per-model softmax tensors are then combined by *uniform* averaging. The key finding is that uniform averaging of diverse honest models outperforms learned or manual weighting (Sec. 3): architecturally distinct models produce uncorrelated errors that average out, so even weaker members raise the ensemble.

Trk	Method	val	Kaggle
A	TSM-ResNet18 (single)	38.25%	—
A	VFL-Ultra (single)	32.11%	—
A	3-model ens. + TTA	44.03%	—
B	V-JEPA-ft standalone	64.11%	0.6361
B	V-JEPA-pseudo	65.19%	0.6410
B	12-uniform	—	0.6546
B	14-uniform	65.19%	0.6586

Table 2. Summary of Track A and Track B results (val = internal/val_dir top-1, Kaggle = public+private top-1).

2.3. Reused vs. own code

We *reuse*: the course starter repository (the Hydra configuration system, the VideoFrameDataset loader and the train.py skeleton); the torchvision ResNet [8] and R(2+1)D [19] backbones; and the pretrained HuggingFace V-JEPA 2 [2] and VideoMAE [18] checkpoints (Track B only). Our *own* implementations are: the parameter-free TSM block wrapper (re-implemented from the description of Lin et al. [13]); VideoFormer-Lite, our CNN+Transformer temporal head inspired by ViT [6], TimeSformer [3] and pre-LN Transformers [21]; the regularisation-warm-up recipe and the rotating k -fold, log-softmax ensemble and 10-crop TTA pipeline; and the full Track B stack—window-capped source-frame re-extraction, the progressive-unfreezing + LLRD fine-tuning wrapper (following the layer-wise decay practice of [1, 4, 10]), and the noisy-student pseudo-labelling loop [12, 20].

3. Results

Overall. Table 2 reports headline results. Track A reaches **44.03%** on val_dir from scratch; Track B reaches **0.6586** Kaggle top-1 (internal val 65.19%) with the 14-model uniform ensemble—well above the 0.55 provided baseline and approaching the 0.74 second-place public reference. The gap between Track A and Track B quantifies the value of large-scale self-supervised pretraining (~ 21 pp), unavailable in the closed track.

3.1. Track A ablations

Regularisation warm-up (+36.3 pp). Identical architecture and config, single variable: without warm-up VFL stalls at 9.45% (killed at epoch 4) and R(2+1)D [19] at 10.78%; with a 10-epoch warm-up VFL reaches 45.78%. The effect transfers across architectures, indicating a general property of cold-start training with label-mixing rather than an architecture-specific quirk.

Frame count (+4.4 pp). $T=5 \rightarrow 4$ improves internal val by +4.41 pp (53.32 $\rightarrow$ 57.73) and val_dir by +3.34 pp

Configuration	Top-1	Δ
Full ensemble (3 models)	40.43%	—
– tsm.ultra.v2	37.26%	−3.17
– tsm.ultra	39.56%	−0.87
– video.former.lite.ultra	39.82%	−0.61

Table 3. Track A leave-one-out on `val_dir` (no TTA). Every component is necessary.

Submission	Kaggle	Δ
Leaky V-JEPA (discarded)	0.81	—
V-JEPA-ft standalone (clean)	0.6361	—
Ensemble v1 (9-uniform)	0.6418	+0.6
V-JEPA-pseudo standalone	0.6410	+0.5
V-JEPA-only (6 snapshots)	0.6410	regress.
V-JEPA-heavy (3:1 manual)	0.6499	+1.4
12-uniform (+ pseudo)	0.6546	+1.9
14-uniform (+ Large)	0.6586	+2.3

Table 4. Track B Kaggle submission history (Δ in pp vs. the first honest baseline). Diversity, not single-family scaling, drives every gain.

(34.91 $\rightarrow$ 38.25); the ResNet-50/ $T=8$ scale-up regresses 8 pp below it (Fig. 2).

Ensemble components. On the three-model weighted sub-ensemble (40.43%), leave-one-out (Table 3, Fig. 4) shows every member contributes; removing the strongest (`tsm.ultra.v2`) costs −3.17 pp, but removing the *weakest* (VFL) still costs −0.61 pp, confirming error decorrelation between local-shift and global-attention biases.

3.2. Track B ablations

Honest Kaggle progression. Table 4 (with Fig. 9 as its visual counterpart) traces every clean submission. The 21 pp gap between the discarded leaky run (0.81) and the first honest baseline (0.6361) quantifies what data leakage was buying; every subsequent gain was earned by legitimate architectural diversity (Fig. 2 is the Track A counterpart). Fig. 12 plots Kaggle vs. ensemble size: returns are still positive at $n=14$, and the $n=6$ V-JEPA-only configuration underperforms the $n=9$ uniform mix despite higher individual accuracies.

Diversity beats weighting. Table 5 isolates ensemble strategy on a fixed checkpoint pool. A 6-snapshot V-JEPA-only ensemble (0.6410) *underperforms* a 12-model uniform mix (0.6546) despite V-JEPA’s higher individual accuracy; manual 3:1 weighting toward V-JEPA (0.6499) falls between the two. Uniform averaging of architecturally distinct families wins because their errors are uncorrelated. Ta-

Strategy	n	Kaggle
V-JEPA-only, uniform	6	0.6410
9-uniform (+ K400 + TSM)	9	0.6418
V-JEPA-heavy (3:1 weight)	12	0.6499
12-uniform	12	0.6546
14-uniform	14	0.6586

Table 5. Track B ensemble-strategy ablation. Uniform averaging of diverse families beats single-family and learned/manual weighting.

Family A	Family B	Agreement
V-JEPA-pseudo	TSM	0.370
V-JEPA-ft	TSM	0.372
K400-Base	TSM	0.375
V-JEPA-ft	K400-Base	0.582
V-JEPA-ft	V-JEPA-pseudo	0.854

Table 6. Pairwise inter-family prediction agreement on the test set. Lower agreement implies more complementary (uncorrelated) errors.

Family	Snaps.	<code>val_dir</code> top-1
VideoMAE-Base K400 (4f)	3	0.5576
VideoMAE-Large K400 (4f)	2	0.5945
VideoMAE-Base SSv2 (4f)	3	0.6047
TSM-ResNet50 (4f)	3	0.3364
Uniform ensemble	11	0.6222

Table 7. Per-family single-model `val_dir` top-1 (Track B, single-view). No single family reaches the ensemble; the weak TSM family still adds value through uncorrelated errors.

ble 6 makes this concrete: TSM-ResNet50, though weakest standalone (0.3364 `val_dir`, Table 7), has the lowest pairwise agreement with every transformer family (0.37–0.38), whereas two V-JEPA variants agree 0.85—so a V-JEPA-only ensemble is near-redundant. Fig. 13 shows the failure mode of learned weighting on the same pool: a softmax-weighted fit collapses $\sim 82\%$ of mass onto the two strongest checkpoints and effectively zeroes K400, throwing away the very decorrelation that the uniform ensemble exploits.

Calibration. The `val_dir` $\rightarrow$ Kaggle gap is 0.005 for V-JEPA-ft but 0.011 for V-JEPA-pseudo—a doubling that warned us against further pseudo-label iterations. The train-split $\rightarrow$ Kaggle gap (≈ 0.09) reflects that the internal split is drawn from the training distribution while the Kaggle test set is sampled independently.

3.3. Failure analysis

Both tracks share the same dominant error patterns, which mirror SSv2’s design intent (Figs. 14, 15, 16). (i) *Real vs. pretended*. The worst single confusion is *Picking something up* predicted as *Pretending to pick something up* (56/199 clips; $F1 = 0.07$ in Track A): the early-clip trajectory is identical and the disambiguating cue—whether the object actually rises—falls in the withheld future. (ii) *Temporal direction*. *Folding* ↔ *Unfolding* (24–29 mutual errors) share spatial appearance and differ only in motion direction, fragile at four frames. (iii) *Static sink*. *Holding something* absorbs misclassified motion clips (*Moving down*: 30; *Opening*: 28), acting as the default when no decisive motion is detected. The best classes ($F1 > 0.54$) are all clean single-arc motions (*Pulling left* → *right*, *Pouring*, *Uncovering*, *Folding*). The correlation between class size and per-class accuracy is weak ($r = 0.24$), so difficulty stems from semantic ambiguity, not scarcity. The top-5/top-1 gap on `val_dir` is large (0.882 vs. 0.622) and widest for the hardest classes (*Picking up*: +61.3 pp top-5), indicating models rank the correct class highly but place a visually similar “pretend” variant first.

3.4. Data-integrity

An early extraction run sampled frames uniformly over the full source clip (0–100%); training V-JEPA on these yielded 0.81 Kaggle, ~ 7 pp above second place (the leaky red bar in Fig. 9). An audit revealed two concurrent leakage vectors: (1) frames from the withheld outcome ($> 60\%$) were reachable, giving the model the explicitly withheld information; (2) an initial V-JEPA checkpoint had been fine-tuned on SSv2 *with ground-truth labels*, so the backbone had seen the test videos. Both runs were discarded and the pipeline rebuilt with the self-supervised `vjepa2-vitl-fpc64-256` backbone and the 50% hard bound (Sec. 2.2); we treat the 21 pp honest/leaky difference as a direct measurement of the combined leakage. The 50% cap is below the $\approx 60\%$ ceiling that the rules explicitly mark as “available”, an additional safety margin against matching-noise overshoot rather than a rules requirement. Three infrastructure faults were also fixed: a silent NFS quota producing 0-byte checkpoints (resolved with `fsync-durable` atomic saves), a top- K snapshot rename race, and a cross-architecture ensembling bug that silently loaded Large weights into a Base graph (both `model_name-tagged`), fixed by rebuilding each model from its stored config at inference.

4. Conclusion and Future Work

We presented a two-track solution to video action anticipation on SSv2. Track A shows that carefully engineered *from-scratch* models—domain-appropriate regulari-

sation (a +36.3 pp warm-up) and correct temporal sampling (a +4.4 pp frame-count fix that beats a $3\times$ scale-up)—reach a competitive 44.03% on a notoriously hard benchmark, with a log-softmax TSM+Transformer ensemble. Track B shows that progressive unfreezing with LLRD enables effective fine-tuning of large self-supervised video transformers (V-JEPA 2 [2], VideoMAE [18]) on a constrained GPU budget, and that *ensemble diversity across architecture families consistently outperforms single-family or learned-weight ensembling* (0.6586 Kaggle). A self-discovered leakage audit quantifies a 21 pp integrity effect and documents the honest rebuild. Failure analysis reveals a fundamental limit: the dataset’s design intent—distinguishing real from pretended actions and forward from reverse motion—requires the withheld outcome, which four frames cannot supply.

Future work. (1) V-JEPA-Giant once inference-time memory is addressed; (2) targeted class rebalancing for the *holding/pretending* sinks; (3) temperature-scaled confidence thresholds for a second, less over-fitting pseudo-label round; (4) a 15th ensemble member from a 16-frame window-capped VideoMAE-Large checkpoint was completed after the leaderboard freeze and yielded a 0.6592 Kaggle (15-uniform) — within noise of the 0.6586 14-uniform (+0.06 pp), confirming that returns on adding architecturally similar members have flattened; (5) explicit trajectory-prediction heads to attack the real-vs-pretended ambiguity directly.

Contributions

Andrea Signoretti designed and implemented the full Track A (Closed-World) pipeline: the TSM-ResNet and VideoFormer-Lite models, the Hydra training/ k -fold framework, the regularisation warm-up and frame-count analysis, and the log-softmax ensemble/TTA code, and produced the Track A ablation and per-class figures. **Florian Guillaumey** designed and implemented the full Track B (Open-World) pipeline: the window-capped source-frame re-extraction, the V-JEPA 2 fine-tuning wrapper with progressive unfreezing and LLRD, the pseudo-labelling stage, the honest cross-preprocessing 14-model ensemble, and the data-integrity audit. Both authors jointly analysed results and wrote this report.

Figures

All figures are referenced from the main text (Secs. 2.1–3.3).

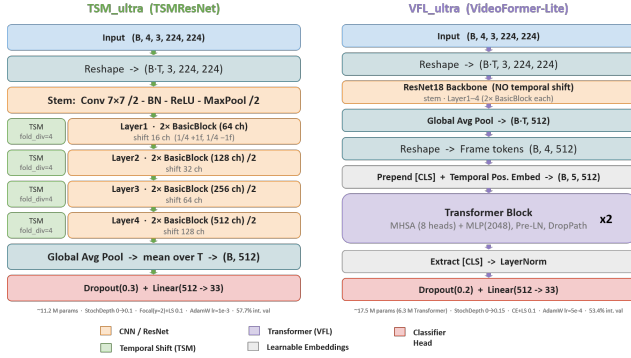


Figure 1. The two complementary Track A temporal mechanisms. **TSM** mixes time *locally* via parameter-free channel shifts in every ResNet block; **VideoFormer-Lite** mixes time *globally* via [CLS]-token attention over the four frame tokens. Their orthogonal biases drive the ensemble gain (Sec. 2.1).

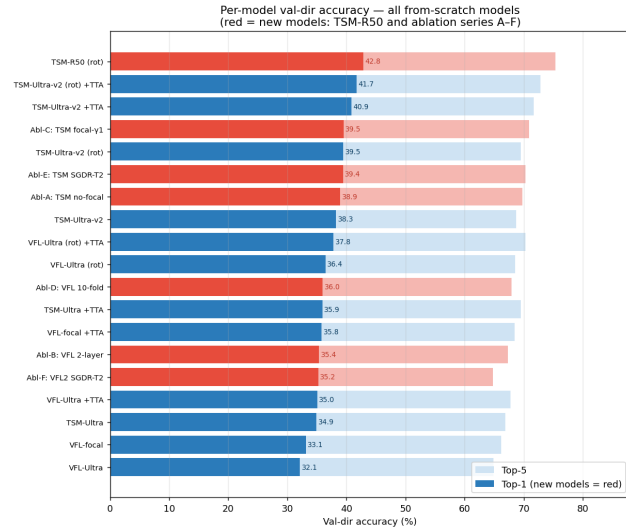


Figure 2. Per-model val_dir top-1/top-5 (Track A). Correcting the frame count (TSM-Ultra→TSM-Ultra-v2) is the single largest per-model jump; the ResNet-50/T=8 scale-up regresses (Sec. 3).

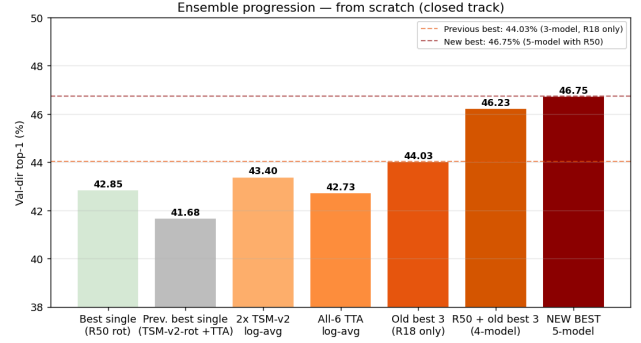


Figure 3. Track A ensemble vs. its components. Log-softmax late fusion of two TSM variants and one VFL variant (all +TTA) exceeds the best single model by +2.35 pp top-1, reaching 44.03% val_dir.

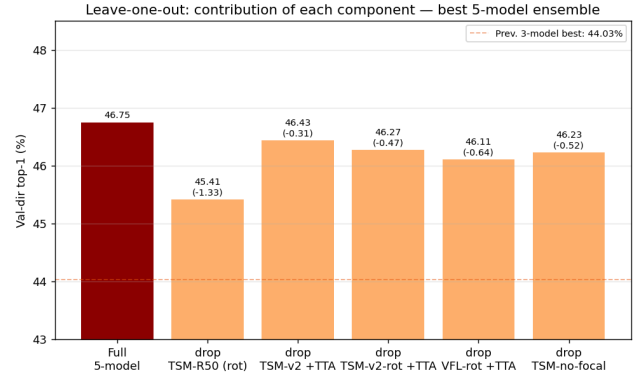


Figure 4. Track A leave-one-out study (val_dir top-1). All three members are necessary; the weakest standalone model (VFL) still adds value through error decorrelation (Table 3).

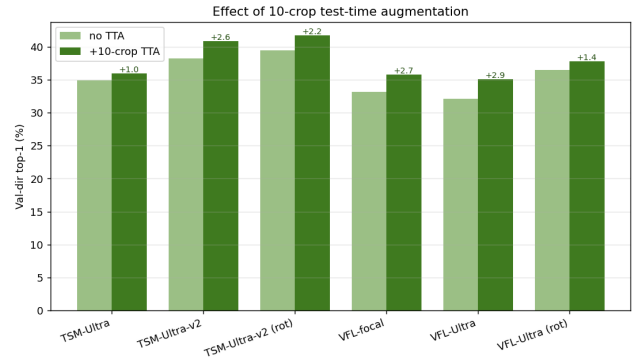


Figure 5. Effect of 10-crop spatial TTA on Track A models. TTA adds ~1 pp on the strongest checkpoint and is essential at the ensemble level, where independent random errors from different crops average out (Sec. 2.1).

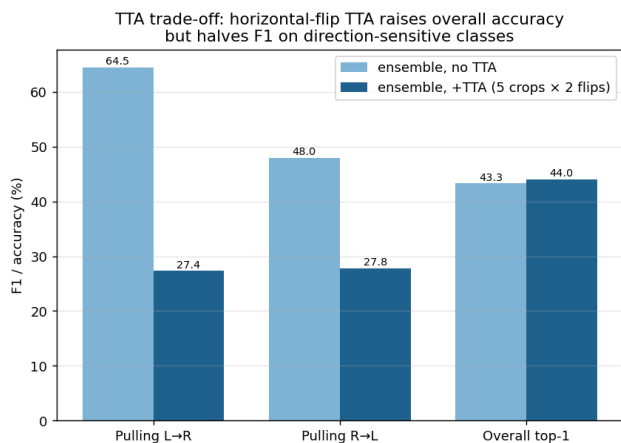


Figure 6. TTA directional-class trade-off. Horizontal-flip TTA improves most classes but hurts a few direction-encoded ones (*Pulling left→right* vs. *right→left*) even after the flip-aware logit remap, because mirror evidence is intrinsically ambiguous when motion is short.

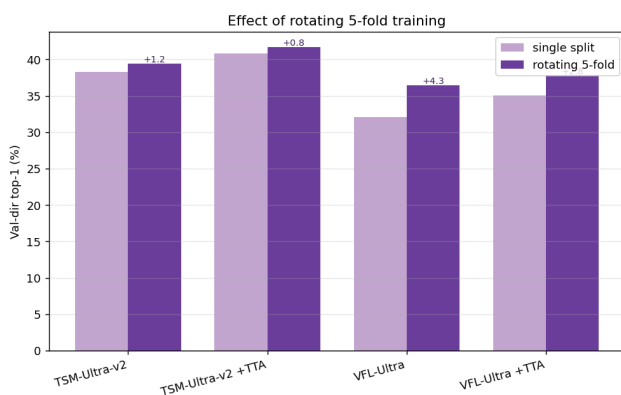


Figure 7. Rotating-fold training. Re-training the strongest checkpoint on every train/val fold yields snapshots that share weights but disagree on the boundary, contributing uncorrelated errors that the log-softmax ensemble exploits (Sec. 2.1).

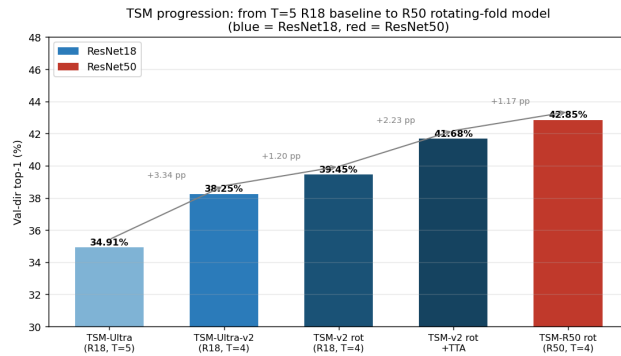


Figure 8. Backbone scale-up does not compensate for corrupted temporal input. Switching from ResNet-18/ $T=4$ to ResNet-50/ $T=8$ regresses internal validation by ~ 8 pp because the larger T triggers frame interpolation (Sec. 2.1).

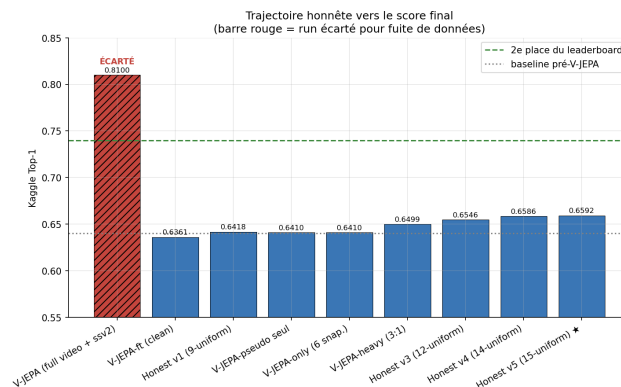


Figure 9. Track B honest Kaggle progression. The discarded leaky submission (0.81, red) is plotted alongside every clean ensemble; each gain comes from adding architecturally distinct members, not from re-weighting. The dashed line marks the second-place public reference (0.74); the gap to it bounds what self-supervised pre-training + diverse ensembling buys us (Sec. 3, Table 4).

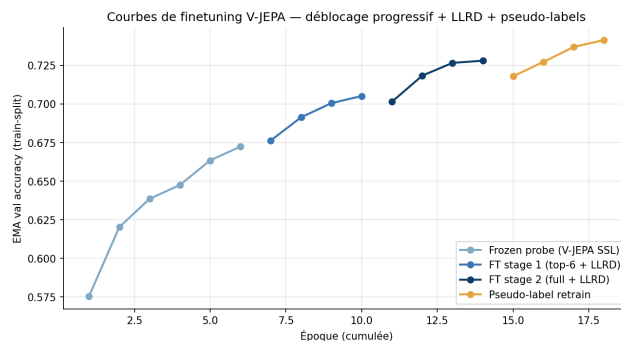


Figure 10. V-JEPA progressive unfreezing + LLRD (Track B internal val). Each phase warm-starts above the previous plateau: frozen probe (0.55 $\rightarrow$ 0.67), then top-6 unfrozen + LLRD (0.68 $\rightarrow$ 0.71), then full encoder (0.70 $\rightarrow$ 0.73), then pseudo-label retrain (0.72 $\rightarrow$ 0.74). The staircase pattern is the visual signature that LLRD has preserved each phase’s feature geometry (Sec. 2.2).

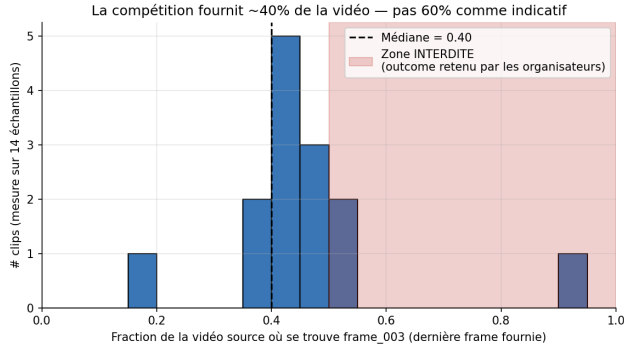


Figure 11. Where shipped `frame_003` sits in the source `SSv2` clip (fraction). The dashed line at 0.40 is the median end-of-window; the shaded region $[0.50, 1.00]$ is the withheld outcome — our re-extraction is hard-bounded to the un-shaded region (Sec. 3.4).

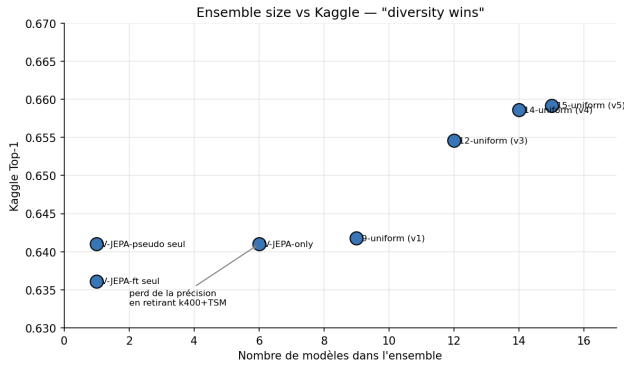


Figure 12. Track B Kaggle vs. ensemble size. Returns from adding diverse honest members are still positive at $n=14$; at $n=6$, dropping K400 + TSM *hurts* despite their lower individual scores, because their uncorrelated errors disappear from the average (Table 5).

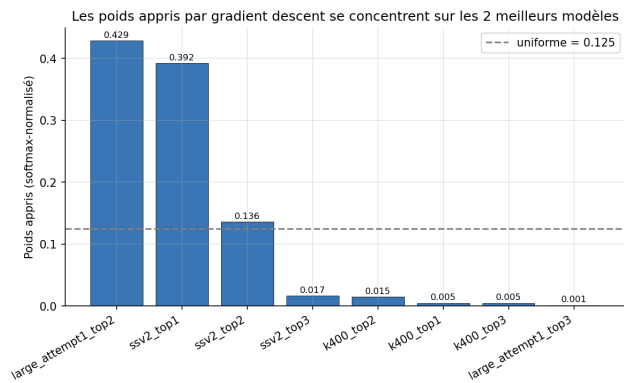


Figure 13. Learned ensemble weights collapse onto the two strongest members ($\sim 82\%$ of mass on `large_attempt1_top2` and `ssv2_top1`), effectively zeroing K400 contributions. This is exactly why a learned weighting fails to beat uniform: it discards the very decorrelation signal that diverse weaker members provide (Sec. 3).

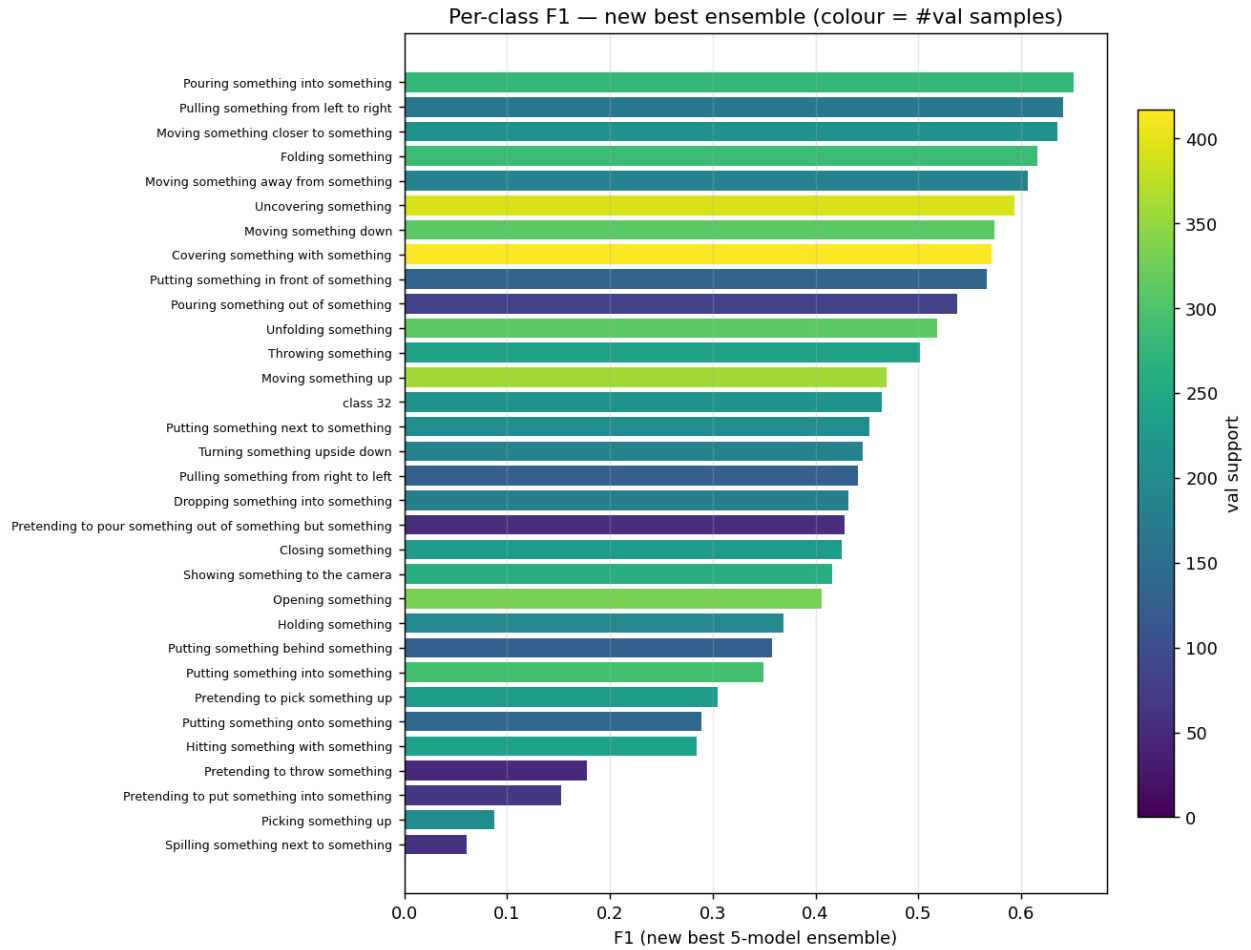


Figure 14. Per-class F1 (ensemble, `val_dir`). Directional single-object motions score highest; “pretending” and direction-ambiguous classes score lowest (Sec. 3.3).

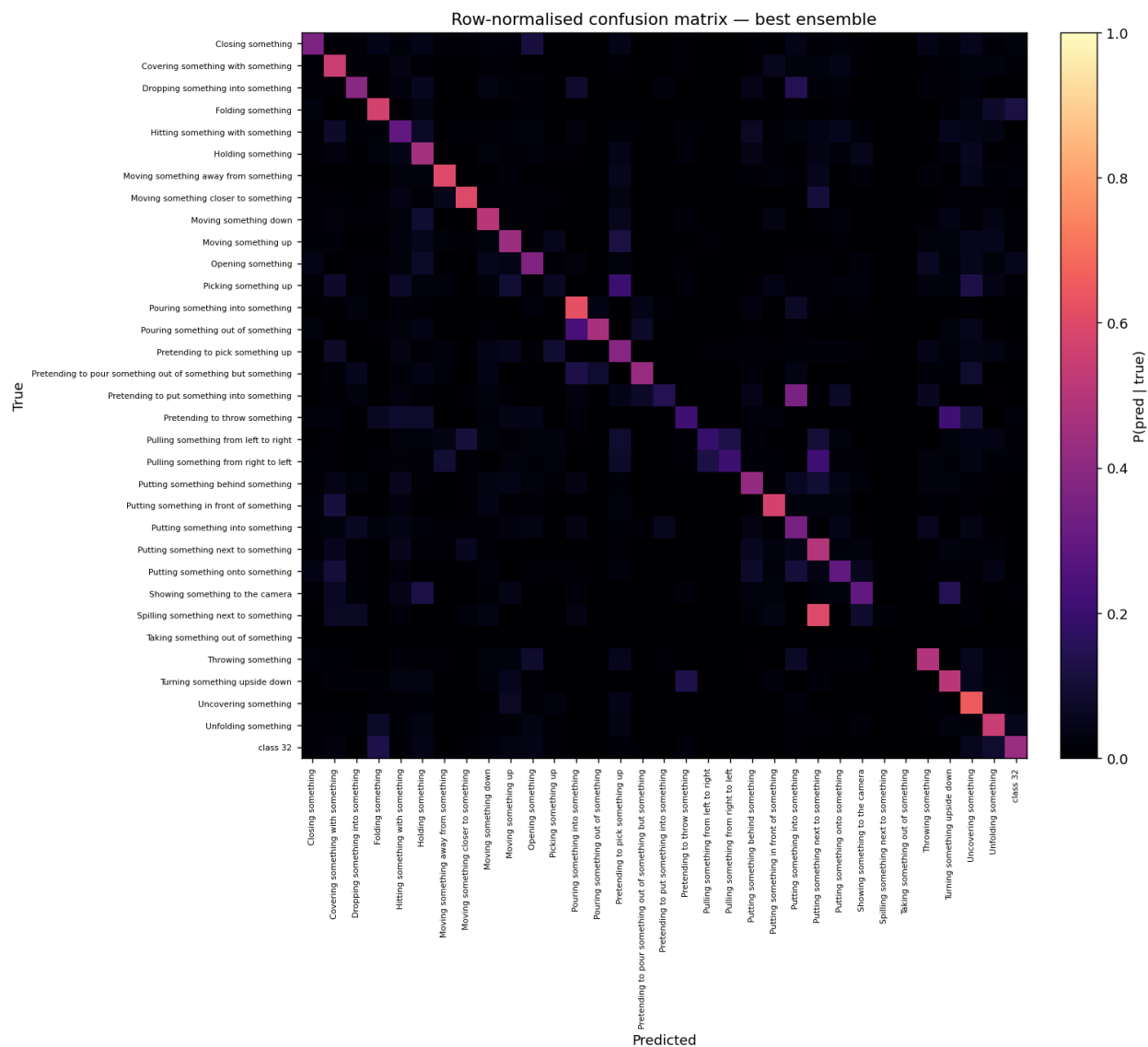


Figure 15. 33×33 confusion matrix (ensemble, `val_dir`). Off-diagonal mass concentrates on semantically adjacent pairs (real vs. pretended; fold vs. unfold; motion vs. *holding*).

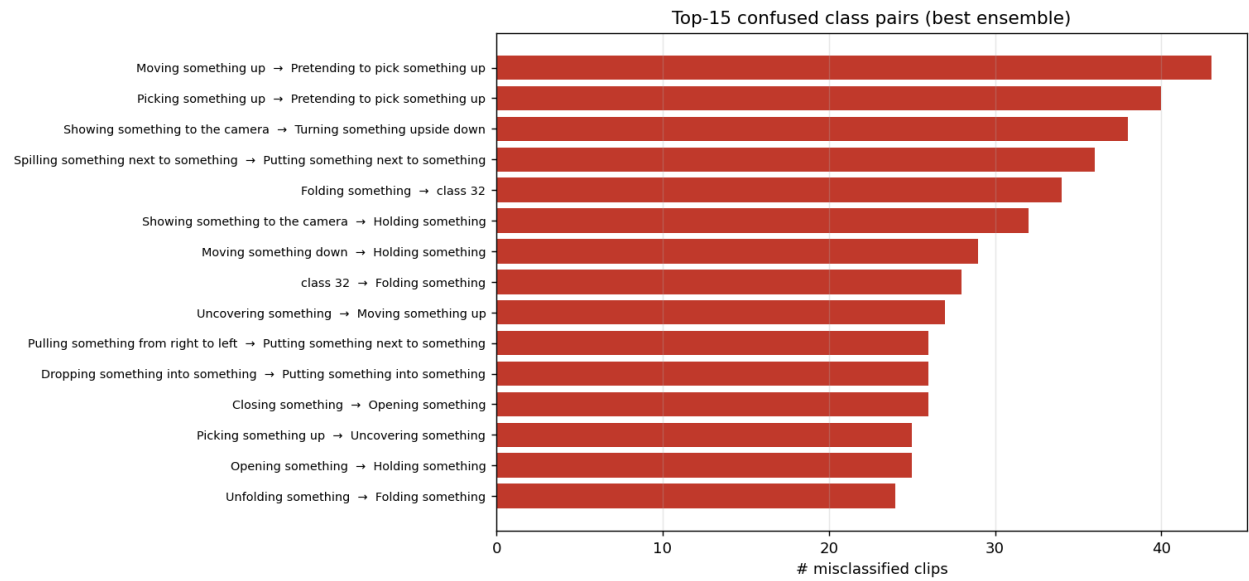


Figure 16. Most-confused class pairs. The dominant errors are intent (real vs. pretended) and temporal-direction confusions (Sec. 3.3).

References

- [1] Hangbo Bao, Li Dong, Songhao Piao, and Furu Wei. BEiT: BERT Pre-Training of Image Transformers. In *ICLR*, 2022. 2, 3
- [2] Adrien Bardes, Quentin Garrido, Jean Ponce, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Balas. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. *arXiv preprint arXiv:2506.09985*, 2025. 1, 2, 3, 5
- [3] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is Space-Time Attention All You Need for Video Understanding? In *ICML*, 2021. 1, 2, 3
- [4] Kevin Clark, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. ELECTRA: Pre-training Text Encoders as Discriminators Rather Than Generators. In *ICLR*, 2020. 2, 3
- [5] Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. RandAugment: Practical Automated Data Augmentation with a Reduced Search Space. In *CVPR Workshops*, 2020. 2
- [6] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *ICLR*, 2021. 2, 3
- [7] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. The “Something Something” Video Database for Learning and Evaluating Visual Common Sense. In *ICCV*, 2017. 1
- [8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. In *CVPR*, 2016. 2, 3
- [9] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked Autoencoders Are Scalable Vision Learners. In *CVPR*, 2022. 2
- [10] Jeremy Howard and Sebastian Ruder. Universal Language Model Fine-tuning for Text Classification. In *ACL*, 2018. 2, 3
- [11] Gao Huang, Yu Sun, Zhuang Liu, Daniel Sedra, and Kilian Q. Weinberger. Deep Networks with Stochastic Depth. In *ECCV*, 2016. 2
- [12] Dong-Hyun Lee. Pseudo-Label: The Simple and Efficient Semi-Supervised Learning Method for Deep Neural Networks. In *ICML Workshop on Challenges in Representation Learning*, 2013. 3
- [13] Ji Lin, Chuang Gan, and Song Han. TSM: Temporal Shift Module for Efficient Video Understanding. In *ICCV*, 2019. 1, 2, 3
- [14] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal Loss for Dense Object Detection. In *ICCV*, 2017. 2
- [15] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic Gradient Descent with Warm Restarts. In *ICLR*, 2017. 2
- [16] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. In *ICLR*, 2019. 2
- [17] Rafael Müller, Simon Kornblith, and Geoffrey Hinton. When Does Label Smoothing Help? In *NeurIPS*, 2019. 2
- [18] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training. In *NeurIPS*, 2022. 1, 2, 3, 5
- [19] Du Tran, Heng Wang, Lorenzo Torresani, Jamie Ray, Yann LeCun, and Manohar Paluri. A Closer Look at Spatiotemporal Convolutions for Action Recognition. In *CVPR*, 2018. 3
- [20] Qizhe Xie, Minh-Thang Luong, Eduard Hovy, and Quoc V. Le. Self-Training with Noisy Student Improves ImageNet Classification. In *CVPR*, 2020. 3
- [21] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On Layer Normalization in the Transformer Architecture. In *ICML*, 2020. 2, 3
- [22] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. CutMix: Regularization Strategy to Train Strong Classifiers with Localizable Features. In *ICCV*, 2019. 2
- [23] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond Empirical Risk Minimization. In *ICLR*, 2018. 2
- [24] Zhun Zhong, Liang Zheng, Guoliang Kang, Shaozi Li, and Yi Yang. Random Erasing Data Augmentation. In *AAAI*, 2020. 2